

Trailer Controller Wiring Reference

System overview

Battery -> Disconnect -> Main Fuse -> Positive Bus -> 12-way Fuse Block. Battery negative -> Negative Bus. Fuse block feeds lighting, buck converter, and winch control coils. Winch motor power bypasses the accessory fuse block and is switched only by the reversing contactor.

Fuse	Circuit	Rating
F1	Arduino + Buck	3A
F2	Light 1	3A
F3	Light 2	3A
F4	Light 3	3A
F5	Light 4	3A
F6	LED Strips Branch A	10A
F7	LED Strips Branch B	7.5A
F8	Dual LED Pods	5A
F9	Winch UP Coil	5A
F10	Winch DOWN Coil	5A
F11	Spare	10A
F12	Spare	10A

Arduino I/O

Inputs 22-33: lighting buttons and winch buttons.

Outputs 34-41: four pod lights, LED strip circuit, dual-pod circuit, winch UP, winch DOWN.

A0: battery voltage monitor.

Notes

MOSFET modules are low-side switches. LED strip circuit uses one control output but two fused power branches. Safe startup forces all outputs OFF. Winch UP and DOWN outputs are software interlocked.